

# CFD 2025-26 – Lab 11 – solution

## Exercise 1

- 1 We consider the two problems separately, and we set up a fixed-point method that starts from an initial guess and then iteratively updates velocity, pressure and temperature by solving two linear problems, of the following form:

Given  $\mathbf{u}_0, p_0, T_0$ , for any  $n = 1, 2, \dots$  find  $\mathbf{u}_k, p_k, T_k$  such that

$$\begin{aligned} (1) \quad & \begin{cases} -\nu \Delta \mathbf{u}_k + (\mathbf{u}_* \cdot \nabla) \hat{\mathbf{u}}_* + \nabla p_k + \beta \mathbf{g}(T_{**} - T_{\text{ref}}) = 0, & \text{in } \Omega, \\ \operatorname{div} \mathbf{u}_k = 0, & \text{in } \Omega, \end{cases} \\ (2) \quad & -k \Delta T_k + \mathbf{u}_{***} \cdot \nabla \hat{T}_{***} = 0, \quad \text{in } \Omega, \end{aligned}$$

where the linear terms depend on the solution  $\mathbf{u}_k, p_k, T_k$  to find at the current iteration, while different choices are possible for the other terms, which are nonlinear either because they are the product of components of the solution with itself (the two advection terms), or because of the dependence of  $T$  on  $\mathbf{u}$  due to the coupling (the buoyancy term  $\beta \mathbf{g}(T - T_{\text{ref}})$ ). The following choice is made:

- $\mathbf{u}_* = \mathbf{u}_{k-1}, \hat{\mathbf{u}}_* = \mathbf{u}_k$ , as usually done;
- $T_{**} = T_{k-1}$  to remove the remaining nonlinearity in the fluid problem;
- $\hat{T}_{***} = T_k$  to preserve the equation type for the thermal problem, namely an advection-diffusion equation in  $T_k$ ;
- $\mathbf{u}_{***} = \mathbf{u}_k$ , since once we have solved problem (1), the velocity  $\mathbf{u}_k$  is known and thus problem (2) can be solved as a linear problem.<sup>1</sup>

We now follow the usual steps for the derivation of the weak formulation of differential problems, and we denote by  $\Gamma_{\text{in}}, \Gamma_{\text{out}}, \Gamma_{\text{p}}, \Gamma_{\text{sym}}$  the portion of the boundary  $\partial\Omega$  corresponding to the inlet, outlet, plate, and the rest, respectively. We introduce the spaces

$$\begin{aligned} V &= \{\mathbf{v} \in [H_{\Gamma_{\text{p}}}^1(\Omega)]^2 : \mathbf{u} = \mathbf{g}_U \text{ on } \Gamma_{\text{in}}, \mathbf{u} \cdot \mathbf{n} = 0 \text{ on } \Gamma_{\text{sym}}\}, \\ V_0 &= \{\mathbf{v} \in [H_{\Gamma_{\text{p}} \cup \Gamma_{\text{in}}}^1(\Omega)]^2 : \mathbf{u} \cdot \mathbf{n} = 0 \text{ on } \Gamma_{\text{sym}}\}, \\ Q &= L^2(\Omega), \\ Z &= \{w \in H_{\Gamma_{\text{in}} \cup \Gamma_{\text{p}}}^1(\Omega) : T = g_T \text{ on } \Gamma_{\text{in}} \cup \Gamma_{\text{p}}\}, \\ Z_0 &= H_{\Gamma_{\text{in}} \cup \Gamma_{\text{p}}}^1(\Omega), \end{aligned}$$

where  $\mathbf{g}_U, g_T$  are the Dirichlet data for velocity and temperature, respectively. Introducing the corresponding finite element spaces  $V^h, V_0^h, Q^h, Z^h, Z_0^h$  and denoting by  $\mathcal{R}^h \mathbf{g}_U \in V^h$  and  $\mathcal{R}^h g_T \in Z^h$  the discrete lifting of the Dirichlet data, the fixed-point iteration algorithm for the FE approximation of the problems reads as follows:

<sup>1</sup>Another possibility is to choose  $\mathbf{u}_{k-1}$  and make the two problems completely independent from one another at each single iteration.

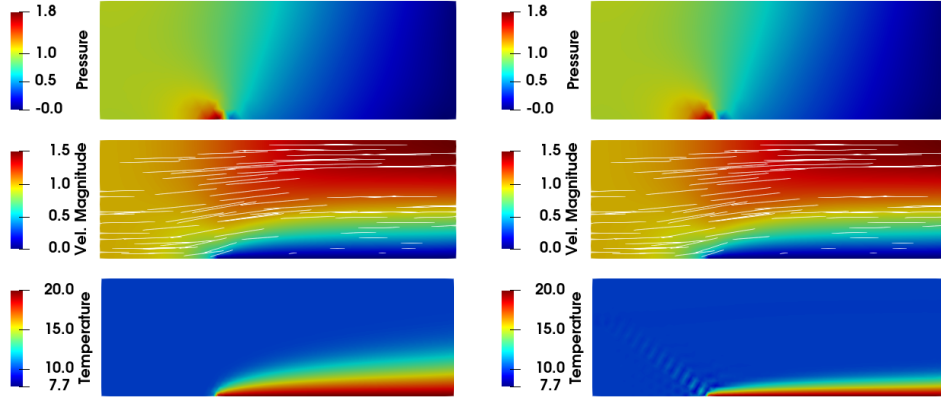


Figure 1: Pressure (top), velocity (middle), and temperature (bottom) for  $k = 10^{-2}$  (left) and  $k = 10^{-3}$  (right).

Given  $\mathbf{u}_0^h \in V^h, T_0^h \in Z^h$ , for any  $n = 1, 2, \dots$  find  $(\hat{\mathbf{u}}_k^h, p_k^h, \hat{T}_k^h) \in V_0^h \times Q^h \times Z_0^h$  such that

$$(1) \quad \begin{aligned} a_U(\hat{\mathbf{u}}_k^h, \mathbf{v}^h; \mathbf{u}_{k-1}^h) - (p_k^h, \operatorname{div} \mathbf{v}^h) - (q^h, \operatorname{div} \hat{\mathbf{u}}_k^h) \\ = -(\beta \mathbf{g}(T_{k-1}^h - T_{\text{ref}}), \mathbf{v}^h) - a_U(\mathcal{R}^h \mathbf{g}_U, \mathbf{v}^h; \mathbf{u}_{k-1}^h) + (q^h, \operatorname{div} \mathcal{R}^h \mathbf{g}_U) \\ \forall (\mathbf{v}^h, q^h) \in V_0^h \times Q^h, \end{aligned}$$

$$(2) \quad a_T(\hat{T}_k^h, w^h; \mathbf{u}_{k-1}^h) = -a_T(\mathcal{R}^h g_T, w^h) \quad \forall w^h \in Z_0^h,$$

where

$$a_U(\mathbf{u}, \mathbf{v}; \mathbf{z}) = (\nu \nabla \mathbf{u}, \nu \nabla \mathbf{v}) + ((\mathbf{z} \cdot \nabla) \mathbf{u}, \mathbf{V}),$$

$$a_T(T, w; \mathbf{z}) = (k \nabla T, \nabla w) + (\mathbf{z} \cdot \nabla T, w).$$

- 2 The code is provided in `CFDlab11.ipynb`. We define different `LinearVariationalProblems` and `LinearVariationalSolvers` for the two problems, and we implement the scheme introduced at the previous point in a while-loop. The choice  $\mathbf{u}_{***} = \mathbf{u}_k$  corresponds to solving the fluid system before the thermal system in the loop.

The fixed-point iteration algorithm converges in 5 iterations to the solution displayed in Figure 1, left. We can observe the thermal layer developing from the plate due to the diffusion of its high temperature and the transport by the flow. We also notice the velocity profile transitioning from flat at the inlet to (half-)parabolic at the outlet, due to the boundary condition on the bottom wall passing from symmetry to no slip.

- 3 The number of fixed-point iterations for different values of  $k$  are reported in the following table:

| $k$                    | $10^{-2}$ | $10^{-3}$ | $10^{-4}$ |
|------------------------|-----------|-----------|-----------|
| Fixed-point iterations | 5         | 5         | 5         |

We do not observe differences in the number of iterations, although the temperature distribution shows significant spurious oscillations for low diffusivity, as shown in Figure 2, left. This independence of the number of iterations from  $k$  is consistent with the temperature distribution affecting relatively little the flow distribution.

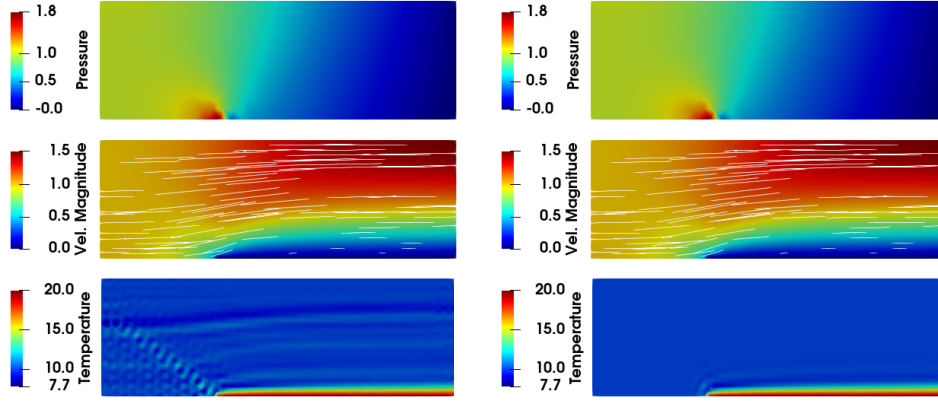


Figure 2: Pressure (top), velocity (middle), and temperature (bottom) for  $k = 10^{-4}$ , without (left) and with stabilization (right).

- 4 The spurious oscillations observed in the previous point can be related to the fact that, with small  $k$ , the thermal problem becomes an advection-dominated problem. As suggested from the numerical analysis of such problems, we can address this issue by introducing a SUPG stabilization. Since the velocity and pressure distributions do not present any numerical artifacts, we stabilize **only the thermal problem**, whose formulation becomes

$$a_T(\hat{T}_k^h, w^h; \mathbf{u}_{k-1}^h) + \sum_{K \in \mathcal{T}_h} \delta \frac{h_K}{\bar{u}_K} \int_K (L_k T_k^h)(L_k^{\text{SS}} w^h) d\mathbf{x} = -a_T(\mathcal{R}^h g_T, w^h) \quad \forall w^h \in Z_0^h$$

where  $\bar{u}_K = \frac{1}{K} \int_K \mathbf{u}_{k-1}^h d\mathbf{x}$  is the average value of the advection velocity on an element  $K$  of the mesh, and  $L_k$  and  $L_k^{\text{SS}}$  are the differential operator associated to the equation and its skew-symmetric component, respectively, namely

$$L_k T_k^h = -k \Delta T_k^h + \mathbf{u}_{k-1}^h \cdot \nabla T_k^h, \quad L_k^{\text{SS}} w^h = \mathbf{u}_{k-1}^h \cdot \nabla w^h + \frac{1}{2} \operatorname{div}(\mathbf{u}_{k-1}^h) w^h.$$

Notice the skew-symmetry correction in  $L_k^{\text{SS}}$ , involving the divergence of the advection field (cf. also the lab on the SUPG stabilization of Navier-Stokes equation).

The number of iterations of the fixed-point algorithm is reported in the following table:

| $k$                    | $10^{-2}$ | $10^{-3}$ | $10^{-4}$ |
|------------------------|-----------|-----------|-----------|
| Fixed-point iterations | 5         | 5         | 5         |

Once again, we do not observe a significant dependence of such number from  $k$ .

The results of the stabilized version of the problem, for  $k = 10^{-4}$ , are displayed in Figure 2, right. We can notice how the spurious oscillations are not present anymore. The temperature distribution is not completely smooth, but this is consistent with the physics of the problem: a reduced thermal diffusivity results in a less smooth temperature distribution.

**Post-processing.** The figures reported here can be reproduced using the Paraview state uploaded on WeBeep: in Paraview, go for *File*  $\rightarrow$  *Load State*  $\rightarrow$  select the file **lab11paraviewState.pvsm** you have downloaded  $\rightarrow$  *Search files under specified directory*  $\rightarrow$  select the directory in which your pvd/vtu files are.